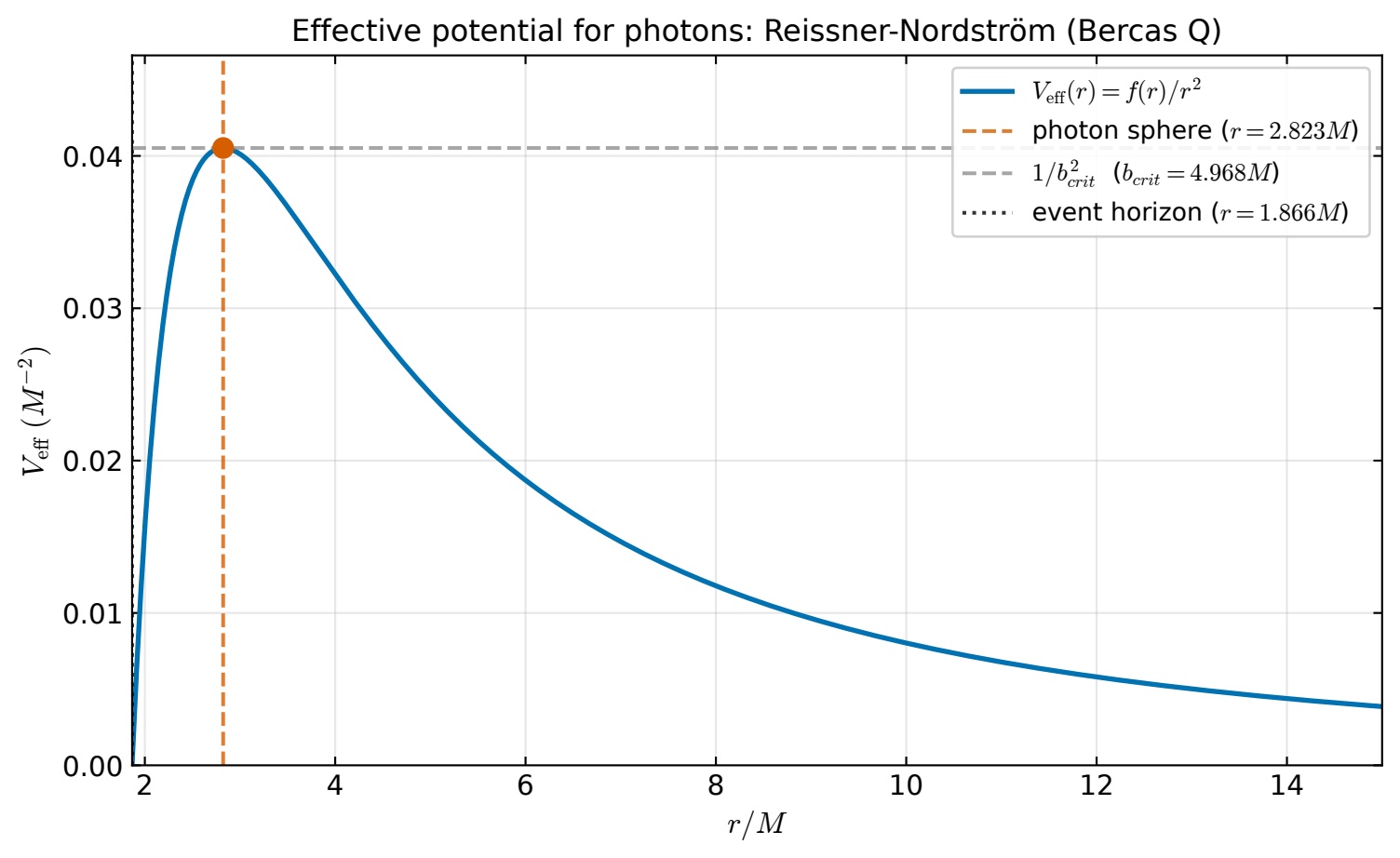
Effective potential for photons: Reissner-Nordström (Bercas Q)
$V_{\rm eff}(r) = f(r)/r^2$
photon sphere ($r = 2.823M$)
$1/b_{crit}^2$ ($b_{crit} = 4.968M$)
event horizon ($r = 1.866M$)
$V_{\rm eff}\ (M^{-2})$
r/M
0.00
0.01
0.02
0.03
0.04
2
4
6
8
10
12
14